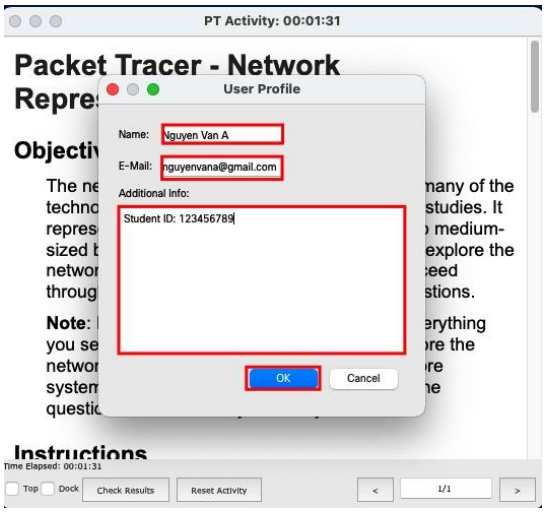
3.5.2.4 Packet Tracer – Learn to use Packet Tracer

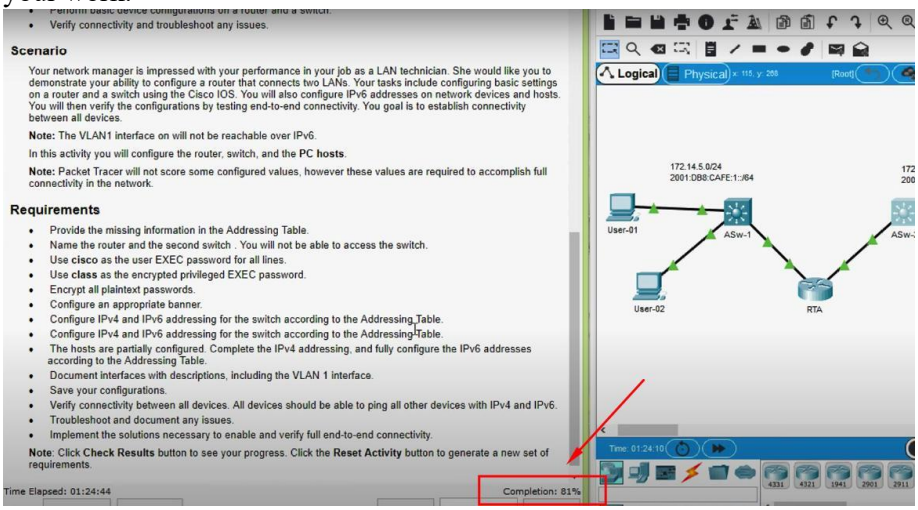
YouTube Link: https://www.youtube.com/watch?v=97UzJKOGmkk&ab_channel=Mr.Mckee

Notes:

1. Remember to include your Name – Email – and Student ID inside the User Profile box



2. After finishing, please take a Packet Tracer screenshot which indicates the completion percentage as evidence for your work.



3. Please submit both .pdf and .pka on Blackboard before the deadline.

Objectives

- Develop an understanding of the basic functions of Packet Tracer.
- Create/model a simple Ethernet network using 3 hosts and a switch.

- Observe traffic behavior on the network.
- Observer data flow of ARP broadcasts and pings.

Step 1: Create a logical network diagram with 3 PCs and a switch.

During an activity, to ensure that the instructions always remain visible, click the “top” check box in the instruction box window.

The bottom left hand corner of the Packet tracer screen displays the icons that represent device categories or groups, such as Routers, Switches, or End Devices.

Moving the cursor over the device categories will show the name of the category in the box. To select a device, first select the device category. When the device category is selected, the options within that category appear in the box next to the category listings. Select the device option that is required.

a. Select **End Devices** from the options in the bottom left-hand corner. Drag and drop 3 generic PCs onto your design area.

b. Select **Switch** from the options in the bottom left-hand corner. Add a 2960 switch to your prototype network by dragging it onto your design area.

c. Select **Connections** from the bottom left-hand corner. Choose a copper straight-through cable type. Click the first host (PC0) and assign the cable to the **FastEthernet0** connector. Click the switch (Switch0) and select a connection FastEthernet0/1 for PC0.

d. Repeat step c for PC1 and PC2. Select FastEthernet0/2 on the Switch0 for PC1 and FastEthernet0/3 for PC2.

e. There should be green dots at both ends of each cable connection after the network has converged. If not, double check the cable type selected.

Step 2: Configure host names and IP addresses on the PCs.

a. Click **PC0**.

b. Select the **Config** tab. Change the PC Display Name to **PC-A**. Select **FastEthernet** tab on the left and add **192.168.1.1** as the IP address and **255.255.255.0** as the subnet mask. Close the PC-A configuration window by selecting the x in the upper right hand corner.

c. Click **PC1**.

d. Select the **Config** tab. Change the PC Display Name to **PC-B**. Select **FastEthernet** tab on the left and add **192.168.1.2** as the IP address and **255.255.255.0** as the subnet mask. Close the PC-B configuration window.

e. Click **PC2**.

f. Select the **Config** tab. Change the PC Display Name to **PC-C**. Select **FastEthernet** tab on the left and add **192.168.1.3** as the IP address and **255.255.255.0** as the subnet mask. Close the PC-C configuration window.

Step 3: Observe the flow of data from PC-A to PC-C by creating network traffic.

a. Switch to **Simulation Mode** by selecting the tab that is partially hidden behind the Real Time tab in the bottom right-hand corner. The tab has the icon of a stopwatch on it.

b. Click **Edit Filter** in the Edit List Filter area. In the event list filter, click **All/None** to deselect every filter. Click **Edit Filter**. Select **ARP** and **ICMP** filters under IPv4 tab.

c. Select a Simple PDU by clicking the closed envelope on the right vertical toolbar. Move your cursor to the display area of your screen. Click on PC-A to establish the source. Move your cursor to PC-C and click to establish the destination.

Note: Notice that two envelopes are now positioned beside PC-A. One envelop is ICMP, while the other is ARP. The Event List in the Simulation Panel will identify exactly which envelop represents ICMP and which represents ARP.

d. Select **Auto Capture / Play** from the Play Controls area of the Simulation Panel. Below the Auto / Capture button is a horizontal bar, with a vertical button that controls the speed of the simulation. Dragging the button to the right will speed up the simulation, while dragging is to the left will slow down the simulation.

e. Observe the path ICMP and ARP envelope. Click **View Previous Event** to continue when the buffer is full.

f. Click **Reset Simulation** in the Simulation Panel. Notice that the ARP envelop is no longer present. This has reset the simulation but has not cleared any configuration changes or dynamic table entries, such as ARP table entries. The ARP request is not necessary to complete the ping because PC-A already has the MAC address in the ARP table.

g. Click **Capture / Forward**. The ICMP envelop will moved from the source to the switch and stop. The **Capture / Forward** allows you to move the simulation one step at a time. Continue selecting the **Capture/Forward** until you complete the event.

h. Choose the Power Cycle Device button on the bottom left, above the device icons.

i. An error message will appear asking you to confirm reset. Click **Yes**. Now both the ICMP and ARP envelops are present again. The power cycle will clear any configuration changes not saved and will clear all dynamic table entries, such as the ARP and MAC table entries.

j. Exit the simulation mode and allow the network to converge.

k. After the network has converged, enter the simulation mode.

Step 4: View ARP Tables on each PC.

a. Click **Auto Capture / Play** to repopulate the ARP table on the PCs. Click **View Previous Event** when the buffer is full.

b. Select the magnifying glass on the right vertical tool bar.

c. Click **PC-A**. The ARP table for PC-A will appear. Notice that PC-A has an ARP entry for PC-C. View the ARP tables for PC-B and PC-C as well. Close all ARP table windows.

d. Click the **select** tool on the right vertical tool bar. (this is the first icon present in the toolbar)

e. Click PC-A and select the **Desktop** tab.

f. Select the **Command Prompt** and type the command **arp -a** and press enter to view the ARP table from the desktop view. Close the PC-A configuration window.

g. Examine the ARP tables for PC-B and PC-C.

h. Close the **Command Prompt** window.

i. Click check results to verify that the topology is correct.

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